MIL-T-36503A
6 March 1974
SUPERSEDING
MIL-T-36503
August 18, 1966

MILITARY SPECIFICATION

TRACTION APPLIANCE, CERVICAL INJURY

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers traction appliances, suitable for providing constant traction on a patient suffering from a cervical spinal or lower extremity injury.

1.2 Classification.

Type I - For use with Stryker Frame and Foster Bed.
Type II - For use with litter.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

V-T-276	Thread, Cotton.
QQ-A-200/9	Aluminum Alloy Bar, Rod, Shapes, Tube, and Wire, Extruded 6063
QQ-A-225/6	Aluminum Alloy Bar, Rod, and Wire, Rolled, Drawn, or Cold Finished, 2024.
QQ-A-225/8	Aluminum Alloy, Bar, Rod, Wire, and Special shapes, Rolled, Drawn or Cold Finished, 6061.

FSC 6530

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SPECIFICATIONS

Federal

QQ-A-250/11	Aluminum Alloy 6061, Plate and Sheet.
QQ-S-763	Steel Bars, Shapes And Forgings, Corrosion-Resisting.
WW-T-700/6	Tube, Aluminum Alloy, Drawn, Seamless, 6061.
CCC-C-419	Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered.
CCC-D-950	Dyeing and Aftertreating Processes for Cotton Cloths.
PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Box, Wood, Cleated Plywood.
PPP-B-621	Box, Wood, Nailed and Lock Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

SPECIFICATIONS

Military

MIL-W-530	Webbing, Textile, Cotton, General Purpose, Natural or in Colors.
MIL-C-2520	Twine, Linen (Waxed and Blocking).
MIL-T-3530	Thread and Twine, Mildew Resistant or Water Repellent Treated.
MIL-W-4088	Webbing, Textile, Woven Nylon.
MIL-T-5038	Tape, Textile and Webbing, Textile, Reinforcing, Nylon.
MIL-C-5651	Cord, Elastic, Exerciser and Shock Absorber for Aeronautical Use.
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water Vapor-proof or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-66	Steel: Chemical Composition and Hardenability.
FED-STD-751	Stitch, Seam and Stitching.

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWINGS

Defense Personnel Support Center (DPSC)

24043	Traction Device, Elastic Cord, Portable, Collins, Litter type.
24044	Traction Device, Elastic Cord, Portable, Collins, Stryker and Foster Bed Type.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS.

3.1 Materials. All materials used in fabrication of the traction appliances shall be as specified herein:

3.1.1 Aluminum Alloy.

3.1.1.1 Tubing. Tubing shall conform to WW-T-700/6 for grade 6061, temper T6.

* 3.1.1.2 Bars and Shapes. Aluminum bars shall conform to QQ-A-225/6 for alloy 2024, temper T3510, or QQ-A-225/8 for alloy 6061, temper T6. Extruded aluminum channel and angles shall conform to QQ-A-200/9 for Alloy 6063, temper T5.

* 3.1.1.3 Plate and Sheet. Aluminum plate and sheet shall conform to QQ-A-250/11 for alloy 6061, temper T6.

3.1.2 Corrosion-Resisting Steel. Corrosion-resisting steel sheets, tubing, and bars and rods shall be type AISI 304 conforming to the chemical requirements in FED-STD-66. Bars and rods shall conform to QQ-S-763. Steel tubing shall be seamless or continuously butt-welded and shall conform to the chemical requirements of FED-STD-66 for AISI 304.

3.1.3 Carbon steel. Cold-rolled carbon steel shall be AISI 1010 or higher-carbon-content steel in conformance with FED-STD-66.

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3.1.4 Thread. Needle and bobbin thread shall be 4 ply cotton, ticket #12, and vat dyed Olive Drab Number 7 in conformance with V-T-276. Needle thread shall be type 1B3, and bobbin thread shall be type 1A3.

3.1.5 Cloth. Cloth shall be Number 10 cotton duck vat dyed Olive Drab Number 7 in conformance with type I of CCC-C-419.

3.1.6 Webbing. Cotton webbing shall be vat dyed Olive Drab Number 7 and shall conform to type IIB, class 4 of MIL-W-530. Nylon webbing shall be dyed Olive Drab number 613, condition "R" and shall conform to type II of MIL-W-4088. Nylon reinforcing (binding) tape and webbing shall conform to type III of MIL-T-5038.

3.1.7 Cord. Elastic cord shall be in conformance with MIL-C-5651. Waxed linen cord shall be 12 ply and type I in accordance with MIL-C-2520.

3.2 Construction.

3.2.1 Type I. The type I traction appliance shall consist of four subassemblies described as the traction cable carriage, carriage adapter, tension adjustment, tension indicator with traction cord, and a satchel type carrying case, and shall be in accordance with Defense Personnel Support Center (DPSC), Drawing No. 24044.

3.2.1.1 Carrying case. The carrying case shall be made of the materials specified in the Bill of Material to the drawing. All stitching shall be type 301 of FED-STD-751. In addition, the cotton duck and threads shall be water repellent and mildew resistant treated in conformance with Type I, Class B of CCC-D-950, and class I of MIL-T-3530, respectively.

* 3.2.2 Type II. The type II traction appliance shall consist of an adapter subassembly, a traction cord, and an "S" hook, and shall be in accordance with DPSC Drawing No. 24043. The adapter assembly shall consist of two metal cups, each provided with a terminal post on which the whipped loop ends of the elastic traction cord shall fit. The subassembly, in turn shall fasten to the handles on one end of the litter referenced in 6.1. The hook shall serve as the means for transferring the traction force to a head-halter or other device for lower extremity injury.

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3.3 Finish. All aluminum components shall be given an anodized finish in conformance with class 2 of MIL-A-8625. Color shall be olive drab.

* 3.4 Identification marking. Each traction appliance shall be legibly and permanently marked with the following:

National Stock Number
Manufacturer's name or registered trademark
Contract or purchase order number
Letters U.S.

Marking secured by adhesives shall not be acceptable.

3.5 Service data. The contractor shall furnish, with each unit, two copies of service data containing complete instructions for installation, operation, maintenance, and a list of component parts. The service data shall include such detailed information which shall permit ready installation and maintenance of the item. All information contained in the manual shall be suitably indexed to permit ready reference to the required information. Diagrams, drawings, and descriptive material shall include adequate information so that the replacement parts can be ordered by the manufacturer's part number, and without confusion as to the item identification. Identical or interchangeable replacement parts shall be identified by a single part number only. The contractor shall mail 15 copies of the service data to the contracting officer for distribution.

3.6 Welding. Joints shall be continuously welded with edges rounded and smooth. Welds shall be durable and sound in every detail. Prior to welding, components shall be properly aligned in their proper positions to insure a straight, rigid, and level assembly.

3.7 Workmanship. Workmanship shall be first class throughout. The traction appliances shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, and auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the traction appliance may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I.

Table I.

	Inspection level	AQL (defects per hundred units)
For visual examination Major defects	II	1.0
Total (major and minor combined) defects	II	4.0
For dimensional examination	S-3	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.

4.3 Examination. The traction appliances shall be examined to determine compliance with all requirements contained in this specification.

* 4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects shown in Table II.

Table II

Categories	Defects*
Major	
101	Unit not complete with all components.
102	Any part damaged or otherwise unuseable.
103	Components not properly aligned or fitted.
104	Welds not continuous, not ground smooth.
105	Exposed edges, corners, and surfaces not free of burs, sharpness.
Minor	
201	Finish not correct color, chipped, cracked, not smooth, not uniform.
202	Identification marking, missing, incomplete, illegible.
203	Service data not supplied in duplicate.

* Examination shall not be restricted to the possible defects listed above.

4.3.2 Dimensional examination. The traction appliances shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One traction appliance, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table III shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table III.

TYPE	Unit package	Intermediate package	Exterior container
I	1 unit	NOT APPLICABLE	4 units
II	1 unit	12 units	48 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package.

5.1.3.1.1 Type I. The traction cable carriage, tension adjustment and tension indicator with traction cord shall be positioned within the carrying case in such a manner as to prevent shifting of or damage to the contents. The carrying case along with carriage adapter shall be packaged in a carton of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to prevent shifting of or damage to the contents. Closure of carton shall be in accordance with PPP-B-636, method 1.

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5.1.3.1.2 Type II. The adapter subassembly, traction cord, (coiled) and "S" hook shall be wrapped in kraft paper and cushioned within a carton of appropriate size constructed in accordance with PPP-B-676, except that commercial colors shall be acceptable. Closure of carton shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2 Intermediate Package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-636. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with paragraph 30.4 of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Special requirements of applicable table shall apply for PPP-B-636 box.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The type I traction appliance is intended for use with NSN 6530-00-680-0501 Turning Frame, Orthopedic Bed and NSN 6530-00-700-8500 Bed, Orthopedic. The type II traction appliance is intended for use with NSN 6530-00-783-7905 Litter, Folding, Rigid Pole, Aluminum Pole.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.2).

* 6.3 This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6530-00-926-4731	TRACTION APPLIANCE, CERVICAL INJURY, Collins, Stryker Frame and Foster Bed Type, Portable
6530-00-926-4732	TRACTION APPLIANCE, CERVICAL INJURY, Collins, Litter Type, Portable

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6.4 This specification does not cover all types, classes, grades, or or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - MD
Navy - MS
Air Force - 03

Review activities:

Army - MD
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Project No. 6530-1285

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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A. GIVE PARAGRAPH NUMBER AND WORDING.

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